# The Path to a Lie: Information Search, Stopping, and Misreporting*

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**Abstract**

This paper examines how people decide when to stop gathering information and how the information they uncover shapes their posterior reports. In an incentivized task in the German Socio-Economic Panel Innovation Sample, 1,283 participants could open ten boxes known to contain each number from 0 to 9 once. They could open as many boxes as they wished but were instructed to report the first value they saw. Their payment equaled the reported value in euros, whether or not the report was truthful. The first opening revealed the truth, and anyone willing to lie could do so without opening another box. Yet many continued looking, and the numbers they encountered shaped how their search unfolded: higher first values discouraged further search, while finding a number higher than any previously seen made participants more likely to stop. Their reports also reflected what they had found, with nearly 90% of those who searched further and reported above the truth choosing a number they had seen, usually the highest one. Together, these findings show the importance that search paths have in shaping what people report.

**Keywords** Lying · Report Selection · Information Acquisition · Survey

**JEL Classification** C91 · D82 · D90

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# 1 Introduction

Many decisions require people to report an outcome after choosing which information to inspect and when to stop gathering it. An auditor may keep sampling transactions until the sample supports signing off on the accounts, while a researcher may examine several econometric specifications before emphasizing the most favorable result (Simmons et al., 2011; Brodeur et al., 2016). In these settings, further search leaves the truth unchanged but can reveal a favorable value that makes a dishonest report easier to justify, giving people a reason to keep looking.

The logic is that lying carries a psychological cost (Gneezy, 2005; Mazar et al., 2008; Abeler et al., 2019; Gerlach et al., 2019), and finding a justification can lower that cost (Shalvi et al., 2015). For example, Shalvi et al. (2011) show that seeing an unrelated favorable outcome can make reporting it feel more acceptable, even when the report is dishonest. These motives can also shape how people gather information. People may avoid information about harmful consequences (Dana et al., 2007; Grossman and van der Weele, 2017; Golman et al., 2017) or choose environments and sources that support favorable interpretations (Saccardo and Serra-Garcia, 2023; Dimant et al., 2024). They may also stop searching more readily when evidence suggests a selfish choice is harmless (Chen and Heese, 2026).

In this paper, we study how and why people continue searching after they already know the truth, and how that search shapes what they ultimately report. The same false report can follow different search paths: one person reports a number they found, while another chooses a number they never saw. Final reports alone hide this distinction and tell us little about how people decided what to report. By observing the truth and the full search path alongside each report, we can examine what encourages further search, what brings it to an end, and how the information encountered shapes reporting.

We use data from an experiment fielded in the 2020 wave of the German Socio-Economic Panel Innovation Sample (SOEP-IS) (Richter and Schupp, 2015). Participants saw ten closed boxes and were told that they contained each value from 0 to 9 exactly once and in random order. They were instructed to report the value in the first box they opened but could inspect additional boxes before reporting. Their payment equaled the reported value, regardless of whether the report was truthful. Crucially, the software

recorded every opening, allowing us to observe the truth, the full sequence of clicks, and the final report.

The values participants uncovered shaped whether they continued searching after the first opening, when they stopped, and what they ultimately reported. A higher first value discouraged further search, while finding a number higher than any previously seen increased the probability of stopping by about 20 percentage points. Their final reports also reflected what they had encountered along the way: among participants who opened additional boxes and misreported, nearly 90% chose a number they had seen, usually the highest one. Our design does not, however, isolate the psychological mechanism behind these patterns. Justification, attention, memory, and searching for a report already in mind may all contribute.

These findings contribute to our understanding of search and dishonest reporting in three ways. First, we extend experiments on favorable counterfactuals (Shalvi et al., 2011; Clist and Hong, 2026) by showing how people's search paths relate to the false reports they make. Closely related, Ay et al. (2024) show that participants search more after low initial outcomes and that dishonest reporters tend to stop at the highest possible outcome. We extend this work by providing causal evidence that favorable encounters shape both stopping and the particular number reported. These findings show that, beyond an initial willingness to search or misreport, what participants encounter along the way matters for their final reports. Second, we bring together research on justification in dishonesty (Mazar et al., 2008; Shalvi et al., 2011; Exley and Kessler, 2024) and motivated information acquisition (Dimant et al., 2024; Saccardo and Serra-Garcia, 2023; Chen and Heese, 2026) by examining how information gathering and reporting unfold together. The design does not establish that participants search for justification, but it shows that favorable values can influence when they stop looking and which number they report. Third, the household-panel sample broadens the evidence on dishonesty beyond the convenience samples commonly used in experiments (Fosgaard, 2020).

More broadly, the experiment highlights why the search behind a report matters. Managers, analysts, and researchers may keep searching until they find a favorable result, then report it while leaving out less favorable evidence encountered along the way. Knowing that the reported result was actually observed does not tell us how it was selected. Exam-

ining the search path, including what else was considered and when the search stopped, can therefore help us understand the final report and its credibility.

The remainder of the paper is organized as follows. Section 2 describes the experiment, and Section 3 discusses why participants might open additional boxes and states three predictions. Section 4 presents the empirical results, Section 5 discusses them, and Section 6 concludes.

## 2 Experimental Design and Data

The survey was conducted as part of the SOEP Innovation Sample, a component of the German Socio-Economic Panel (Richter and Schupp, 2015). Interviews were administered face to face using Computer-Assisted Personal Interviewing (CAPI) between September 2020 and February 2021 (Zweck and Rathje, 2021).[1] Participants completed two incentivized tasks commonly used in the experimental literature on lying, in randomized order. In both tasks, payment depended only on the participant's report. Reporting $r \in \{0, \ldots, 9\}$ paid $r$ euros whether or not the report was truthful. Participants knew that only one task would be payoff-relevant and that a coin flip after both tasks would determine which one they were paid for. The module was offered to 1,603 respondents, of whom 1,330 participated, and 1,318 completed both tasks. Appendix B provides an English rendering of the programmed instructions.

We focus on the task based on the "box paradigm" of Gneezy et al. (2018). Participants saw ten closed black boxes on the survey laptop, each containing a unique number from 0 to 9 in random order. Clicking a box revealed its value. Participants could open as many boxes as they wished but were instructed to remember and report the value in the *first* box they opened. They entered their report privately on the laptop, with the interviewer instructed to leave the room or turn away. The software recorded the full sequence of clicks, including which boxes were opened, in what order, and which values were revealed. Participants were told that the laptop recorded which boxes they opened and how many, but that their survey responses remained anonymous and were not linked to their names.

The other task adapts the private die-rolling paradigm of Fischbacher and Föllmi-Heusi (2013). Participants could roll a fair 10-sided die under a cup as often as they

[1] Despite the COVID-19 pandemic, some SOEP-IS interviews continued in person under strict hygiene protocols.

wished but were instructed to remember and report the result of the *first* roll. They again entered their report privately on the laptop.[2] Because we did not observe the first roll, this task lets us assess whether reports are higher on average than expected from a fair die, but cannot identify individual false reports. After both tasks, participants tossed a coin in view of the interviewer, who recorded which task was selected for payment.

Bosch-Rosa et al. (2025) also use this module to study how privacy and researchers' ability to detect false reports affect dishonesty and concerns about appearing dishonest. They compare the private dice task with the observable box task and examine differences across demographic groups. Our analysis instead uses the full sequence of clicks in the box task, which their study does not examine. Thus, while the dice task is central to Bosch-Rosa et al. (2025), here it serves only as a complementary benchmark for interpreting path-based reporting in the box task.

## 3 Why open more boxes?

Participants know from the start that the boxes contain the numbers 0 to 9 and that they can report any of them. Opening the first box tells them which number they were instructed to report. Therefore, unlike standard search problems, in which search can uncover a better option (e.g., Weitzman, 1979; Choi et al., 2018), further openings change neither the truth nor the available reports. If participants remember the truth and know how they value each report, further search offers no reporting benefit unless seeing another number changes those valuations. If these valuations remain unchanged and opening takes time or effort with no separate benefit, they should stop after the first box.

Seeing a number may, however, make it easier to justify reporting it. Someone who would hesitate to invent a number above the truth may feel more comfortable reporting one they have actually seen. Search may therefore change which number a participant is willing to report, even when the truth is already known. Our model captures this possibility by weighing the benefit of encountering a favorable number against the cost of opening another box. Appendix C presents the assumptions and formal results behind the three predictions below:

[2] Fischbacher and Föllmi-Heusi (2013) use a 6-sided die. We use a 10-sided die to match the 0–9 range of the box task.

**Prediction 1.** *A higher first-box value makes participants less likely to open a second box.*

**Prediction 2.** *Participants are more likely to stop after finding a new maximum.*

**Prediction 3.** *Participants who misreport choose a number they have seen during their search.*

# 4 Results

## 4.1 Descriptives

Of the 1,318 participants who completed both tasks, 1,283 opened at least one box and revealed the value they were instructed to report. We exclude the remaining 35 participants, who submitted a report without opening a box.[3]

Appendix Table A.2 summarizes participants' behavior. Most participants reported truthfully: of the 1,283 participants, 1,152 reported the value in their first box, 21 reported a lower number, and 110 (8.6%) misreported.[4] The predominance of truthful reporting is consistent with evidence that people often forgo profitable opportunities to lie (Abeler et al., 2019) and that dishonesty is less common in representative non-student samples than among students (Fosgaard, 2020).

Participants opened 2.16 boxes on average, with a median of one among truthful reporters and three among those who misreported. Misreporting was more common among participants who continued searching: 96 of the 392 participants who opened additional boxes (24.5%) misreported, compared with 14 of the 891 who opened only one (1.6%). A further 19 participants who searched reported below the truth, giving 115 incorrect reports among searchers. Among the 96 misreporters who opened additional boxes, 86 (89.6%) reported a value they had seen, usually the highest one. Most misreporters also left money on the table: 76 of 110 chose a value below the maximum of 9. Such

[3]The excluded participants reported 4.89 on average, close to the expected value of a randomly opened box, 4.5. Their most common report was 5, chosen by 11 participants. Their reports therefore do not appear unusually high.

[4]We use *misreport* for a report above the truth and *incorrect report* for any report that differs from it.

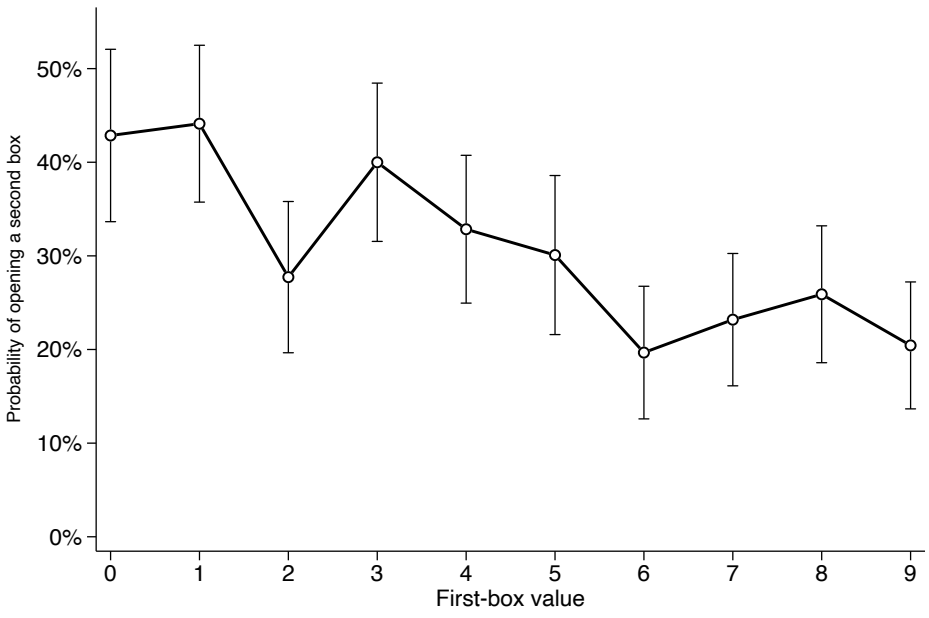
partial misreporting aligns with previous experiments (Fischbacher and Föllmi-Heusi, 2013; Gneezy et al., 2018) and theoretical models (Dufwenberg and Dufwenberg, 2018; Khalmetski and Sliwka, 2019).

### 4.2 Opening a second box

The first value determines what participants earn by reporting truthfully. We begin by examining whether it also affects their decision to search further. Figure 1 plots the first value on the horizontal axis and the probability of opening a second box on the vertical axis. As predicted, participants were generally less likely to open a second box when the first value was higher.

Because the first value was randomly assigned, we can identify its causal effect on further search. Table 1 reports two participant-level regressions: a linear probability model for opening a second box and an OLS regression for the number of distinct boxes opened. Consistent with Figure 1, a one-point increase in the first value reduced the probability of opening a second box by 2.5 percentage points. In this line, higher first values also reduced the total number of distinct boxes opened.

Figure 1: Opening a second box, by first-box value




*Notes:* Points show the empirical share of participants who open a second box at each first-box value. Whiskers show 95% confidence intervals.

Table 1: First-box value and additional openings, with demographic controls

| | Opens second box | | Distinct boxes opened | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| First-box value | -0.025*** | -0.024*** | -0.060*** | -0.057*** |
| | (0.004) | (0.004) | (0.022) | (0.021) |
| Constant | 0.422*** | 0.519*** | 2.437*** | 2.838*** |
| | (0.025) | (0.116) | (0.120) | (0.582) |
| N | 1,283 | 1,283 | 1,283 | 1,283 |
| Task-order control | No | Yes | No | Yes |
| Demographic controls | No | Yes | No | Yes |

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

*Notes:* Participant-level OLS. Columns (1)–(2) are linear probability models for opening a second box; Columns (3)–(4) use the number of distinct boxes opened. Columns (2) and (4) additionally control for randomized task order, gender, age and its square, monthly net household income, education, full-time employment, and residence in East Germany in 1989. Missing demographic covariates are mean-imputed with missingness indicators. Appendix Table A.1 summarizes the individual characteristics. Robust standard errors are in parentheses.

**Result 1** *A higher first-box value makes participants less likely to open a second box and reduces the total number of boxes they open.*

### 4.3 Stopping

For participants who keep looking, we next ask what brings their search to an end. We begin with the 392 participants who opened a second box. Before that opening, they had seen only the truthful value, so holding it fixed allows us to compare responses to a randomly revealed higher or lower number without having to account for other discoveries along the way. Later in the search, a number above the truth may still fall below one already seen. We therefore start with the second opening, where finding a number above the truth and finding a new maximum are the same event, before examining later openings.

Panel A of Table 2 uses three participant-level linear probability models to examine stopping and reporting after the second opening. Column 1 examines whether participants stop, Column 2 whether they ultimately report the second value, and Column 3 whether they misreport. All three models include an indicator for the second value exceeding the truth, first-box fixed effects, and a control for randomized task order.

The results show that a second value above the truth increased the probability of stopping by 20 percentage points relative to one below it. Participants were also 13 percentage points more likely to report the second value when it exceeded the truth.[5] The estimated effect on misreporting was positive, at 6.7 percentage points, but not statistically significant. Therefore, finding a favorable number seems to help participants settle on what to report and bring their search to an end, even though they could have chosen that report without searching further.

We next broaden the stopping analysis to include later openings. A draw sets a *new maximum* when it reveals a number higher than every value the participant has previously seen. To test the model's prediction, we compare stopping after new maxima with stopping after draws below the previous maximum. We retain only openings where

[5] A report can match the second number by chance, even if that number has no influence on what the participant reports. To account for this, we keep each participant's first number and final report fixed and randomly redraw the second number from the nine remaining numbers. We repeat this 100,000 times, checking which reports match and re-estimating the same regression each time. We then check how often the simulated effect is at least as large as the observed effect, in either direction. The observed pattern is unlikely to arise if final reports are unaffected by the second draw ($p < 0.001$).

both types of draw are possible, giving a sample of 1,013 second and subsequent openings from 350 participants.[6]

However, even when both types of draw are possible, the probability of a new maximum varies with the number of unopened boxes and the highest value already seen. Because these factors may also influence stopping, Panel B of Table 2 reports three different linear probability models. Column 1 gives the unadjusted difference in stopping probabilities between new maxima and lower draws. Column 2 uses inverse-probability weights, giving more weight to whichever type of draw was less likely at that opening.[7] Column 3 is unweighted, but includes fixed effects for each combination of opening number and previous maximum, comparing new maxima with lower draws at the same stage of search and with the same previous maximum. It also controls for randomized task order. All three specifications show that participants are more likely to stop after finding a new maximum. Excluding second openings produces similar estimates, showing that the stopping response persists at later stages of information search (Appendix Table A.3).

[6] At the second opening, a new maximum is guaranteed after an initial 0 and impossible after an initial 9. At later openings, other search histories can also make the outcome certain. For example, after seeing 0, 1, and 2, every remaining value exceeds the previous maximum.

[7] Before opening the $k$th distinct box, $11-k$ boxes remain unopened. If the highest previously seen value is $m$, exactly $9-m$ contain a higher value. The probability of a new maximum is therefore $q_{k,m} = (9-m)/(11-k)$. We assign weight $1/q_{k,m}$ to openings that set a new maximum and $1/(1-q_{k,m})$ to those that do not.

Table 2: New maxima, stopping, and report selection

**Panel A. Second opening**

| | Stops after second opening (1) | Final report equals second value (2) | Final report above truth (3) |
|---|---|---|---|
| Second value > truth | 0.200*** | 0.130*** | 0.067 |
| | (0.048) | (0.037) | (0.055) |
| N | 392 | 392 | 392 |
| Task-order control | Yes | Yes | Yes |
| First-box fixed effects | Yes | Yes | Yes |

**Panel B. Stopping at second and later openings**

| | Stop (1) | Stop (2) | Stop (3) |
|---|---|---|---|
| New maximum | 0.178*** | 0.196*** | 0.193*** |
| | (0.027) | (0.032) | (0.030) |
| N | 1,013 | 1,013 | 1,013 |
| Task-order control | No | No | Yes |
| Unadjusted | Yes | No | No |
| Inverse-probability weighted | No | Yes | No |
| Opening-position × prior-maximum fixed effects | No | No | Yes |

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

*Notes:* Linear probability models. Panel A uses participants who opened a second box: Column (1) indicates stopping after that opening, Column (2) indicates a final report equal to the second value, and Column (3) indicates a final report above the truth. Both reporting outcomes include reports submitted after further openings. Panel B uses reached second and later openings at which a new maximum is possible: Column (1) is unadjusted, Column (2) weights each opening by the inverse probability of its realized new-maximum outcome, and Column (3) includes opening-position × prior-maximum fixed effects. All columns include a constant. Panel A includes first-box fixed effects and controls for randomized task order; Column (3) of Panel B also controls for task order. Standard errors are robust in Panel A and clustered by participant in Panel B.

**Result 2** *Finding a new maximum increases the probability of stopping.*

### 4.4 Misreporting

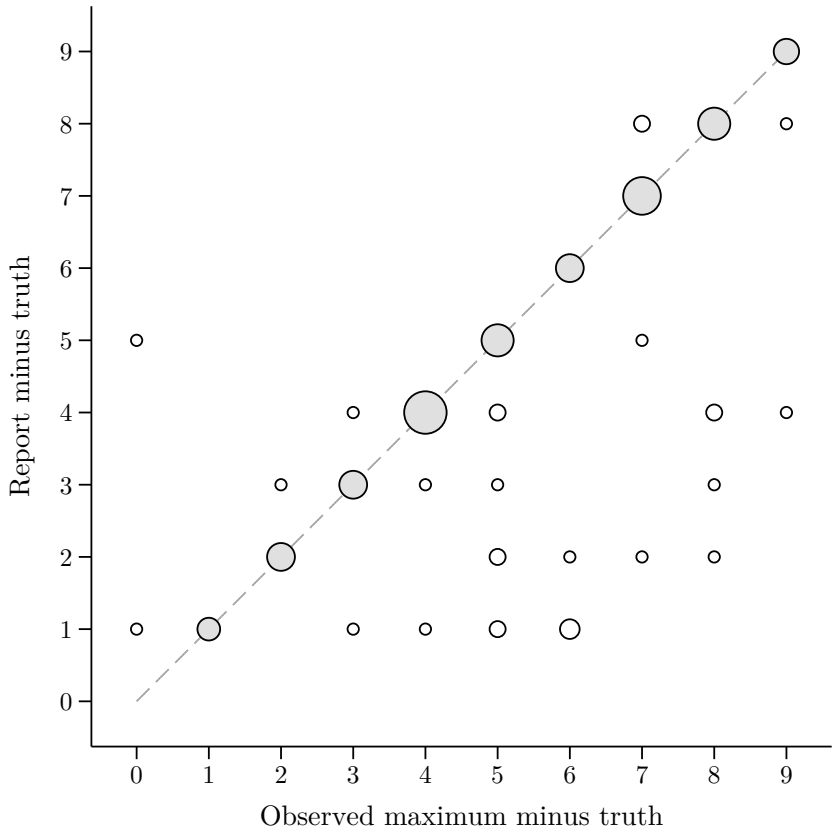
Participants could claim the maximum payment by reporting 9 without ever seeing it, yet most misreporters chose a lower number. We now examine how these reports relate to participants' search paths, focusing on whether they match a value encountered along the way, particularly the highest value seen or the value in the last newly opened box.

Among the 96 participants who opened additional boxes and misreported, 86 chose a value they had seen. Of these, 68 reported the highest value encountered and 18 reported another observed value. The remaining ten reported unseen values, with four choosing a number below their observed maximum and six choosing a number above it. Figure 2 plots the gap between the report and the truth against the gap between the highest observed value and the truth. Most misreports lie on the 45-degree line, where the report equals the highest value seen. Even when they could have claimed a higher payment, participants often selected the highest value they had actually uncovered. The observed maximum may therefore serve as a reference point for misreporting.

Table 3 examines how misreporting varies with the size of the *final improvement*, defined as the increase in the highest value seen when the last new box is opened. Column 1 considers any misreport, while Columns 2 and 3 distinguish misreports matching the last newly revealed value from other misreports. Each outcome equals one for the relevant type of misreport and zero otherwise. The three linear probability models use the same 392 participants who opened additional boxes. All include first-box fixed effects and controls for randomized task order, the number of distinct boxes opened, and the gap between the highest observed value and the truth.

Larger final improvements are associated with more misreporting, but this relationship is concentrated in participants choosing the value in the last newly opened box as their misreport. There is little corresponding association with misreports of other numbers. The pattern remains after accounting for the highest value seen and the number of boxes opened. It therefore connects report selection to what participants discover at the end

Figure 2: Observed evidence and misreports after additional openings




*Notes:* The sample consists of the 96 participants who opened at least two boxes and misreported. Circle area is proportional to the number of participants at each point. Circles filled with light gray indicate reports equal to the highest value seen.

of search, beyond how high a number they encounter overall or how many boxes they inspect.[8]

Table 3: Final improvements and misreporting

| | Any misreport (1) | Last newly opened value (2) | Other misreport (3) |
|---|---|---|---|
| *Final improvement* | 0.052*** | 0.054*** | -0.002 |
| | (0.013) | (0.012) | (0.007) |
| N | 392 | 392 | 392 |
| Task-order control | Yes | Yes | Yes |
| First-box fixed effects | Yes | Yes | Yes |
| Max gap and boxes opened | Yes | Yes | Yes |
| $R^2$ | 0.150 | 0.170 | 0.076 |

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

*Notes:* Participant-level OLS. Column (1) indicates any report above the truth; Column (2) indicates a misreport equal to the last newly opened value; Column (3) indicates every other misreport. The sample includes all participants who opened at least two distinct boxes. All columns include a constant and first-box fixed effects and control for randomized task order, the number of distinct boxes opened, and max gap, the highest observed value minus the truth. Robust standard errors are in parentheses.

These patterns suggest that personally encountered values can provide reference points for misreporting. Participants were free to claim the highest payment without finding the corresponding number, yet most misreporters chose a number they had actually seen.

**Result 3.** *Among participants who open additional boxes and misreport, most choose a value they have seen, usually the highest value encountered.*

## 5 Discussion

Our findings suggest that the numbers encountered during search help shape both stopping decisions and final reports. A favorable number may provide both a reason to stop looking and a report that feels easier to justify. This interpretation is consistent with the predictions in Section 3, in which observing a value higher than the truth makes misreporting easier to self-justify. However, similar patterns could arise without participants consciously searching for a justification. We therefore consider other processes that could connect search, stopping, and reporting.

[8] Appendix Table A.4 includes participants who opened only one box. The association remains positive and similar after controls.

One distinction concerns whether participants settle on a report before or during their search. Some may initially open additional boxes out of curiosity, enjoyment, or uncertainty about the task and only then encounter a favorable number they feel comfortable reporting. Others may already have a report in mind and search until they find it, as in satisficing search (Simon, 1955; Caplin et al., 2011). Either process could lead participants to stop when they find the number they ultimately report. Our data reveal what participants opened and reported, but we cannot distinguish when they decided what their report would be.

Attention and memory offer another explanation for why search paths shape reports. Participants may report a favorable or recent number because they remember it more readily than the truthful first value, without consciously seeking to justify a false report. The benchmarks in Appendix D.2 reject some simple models of recall and reporting errors, but leave room for differences across participants and more flexible forms of motivated recall (Saucet and Villeval, 2019; Carlson et al., 2020). Attention and memory could therefore help shape both the search path and the final report, and our findings do not fully distinguish these processes from deliberate justification.

We also examine whether the stopping and reporting patterns reflect a broader willingness to misreport for money. The separate dice task, adapted from Fischbacher and Föllmi-Heusi (2013), provides an imperfect measure of this willingness. Participants privately roll a ten-sided die numbered from 0 to 9 and report their first roll, with higher reports earning larger payments exactly as in the box task. Because we do not observe the actual roll, a high report may reflect either a high roll or misreporting. Higher dice reports are associated with more misreporting in the box task (Appendix D.1). Participants who report higher dice numbers also respond more strongly to favorable draws during search. Their probability of stopping rises more sharply as the improvement in the highest number seen increases. Together, these patterns are consistent with the idea that search helps participants act on a willingness to misreport. Finding a favorable number may give them a report they can more easily justify and a reason to stop looking.

Participants' willingness to misreport may also depend on how observed or judged they feel. In the panel setting, some may have worried that a dishonest report could affect their relationship with the survey, despite assurances that responses would be evaluated anonymously and not linked to their names. If these concerns discouraged misreporting,

the observed rate may be lower than in otherwise comparable settings where people feel less scrutinized.

Beyond the frequency of misreporting, our findings show how the information encountered during search can shape the content of reports, even when people already know the truth and further search changes neither their reporting options nor the available payments. Future studies could clarify the processes behind this relationship. Keeping the first number visible throughout the task would reduce memory demands and help assess the role of recall. Comparing participants who choose when to stop with others shown the same sequence of numbers could help distinguish the role of control over search from that of the information encountered. Finally, asking participants for a provisional report before further search could help distinguish searching for a number they already intend to report from revising their report in response to new discoveries.

## 6 Conclusion

Reporting is often the final step in a search for information. Before describing an event, offering an assessment, or making a recommendation, people choose which sources to consult, what evidence to examine, and when to stop looking. These decisions shape the information they encounter and may influence what they ultimately communicate. The same report can therefore follow very different search paths. Examining those paths can help us better understand how people arrive at the reports they choose to communicate.

We study how search shapes reporting in an incentivized task embedded in the German Socio-Economic Panel Innovation Sample (SOEP-IS), involving 1,283 participants. Participants were instructed to report the number in the first box they opened but could inspect additional boxes before reporting. They could report any number from 0 to 9, and their payment equaled the reported number in euros, whether or not it was truthful. Further openings therefore provided no new information about the truthful answer or the available payments; they changed only which numbers participants had personally seen. Recording the first number, every subsequent opening, and the final report allows us to trace how search unfolds between learning the truth and deciding what to report.

The results show that the numbers participants encounter shape both their search decisions and their final reports. Participants are less likely to continue searching when the first number is higher and more likely to stop after finding a new maximum. Among

those who open additional boxes and misreport, almost 90% choose a number they have seen, usually the highest one. One interpretation is that participants keep searching after learning the truth because finding a favorable number makes a false report easier to justify. However, favorable or recent numbers may also stand out or be easier to recall; our design cannot fully separate the roles of attention, memory, and justification.

Overall, our findings show that the information people encounter during search can shape what they report. This matters when people compare reviews before recommending a product, consult several sources before describing an event, or try different analyses before presenting a finding. Knowing what they considered and when they stopped helps us trace the path to their final report and judge how much weight to give it.

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# A Extra Tables

Table A.1: Individual characteristics

| | Mean | SD | Min | Max | N |
|---|---|---|---|---|---|
| Female | 0.527 | 0.499 | 0 | 1 | 1318 |
| Age | 54.627 | 18.554 | 17 | 95 | 1318 |
| Education (years) | 12.520 | 2.758 | 7 | 18 | 1279 |
| Net household income (1,000 EUR/month) | 3.242 | 2.124 | 0 | 20 | 1262 |
| Full-time employed (dummy) | 0.308 | 0.462 | 0 | 1 | 1318 |
| Lived in East Germany in 1989 | 0.233 | 0.423 | 0 | 1 | 1316 |
| N | 1318 | | | | |

Table A.2: Descriptive statistics

| | Mean | SD | N |
|---|---|---|---|
| *Full sample* | | | |
| Reported value in box task [0–9] | 4.949 | 2.869 | 1,318 |
| First-box value | 4.614 | 2.880 | 1,283 |
| Report minus first-box value | 0.337 | 1.496 | 1,283 |
| Misreports ($r > x_1$) | 0.086 | 0.280 | 1,283 |
| Boxes opened | 2.161 | 2.273 | 1,283 |
| Maximum value observed | 5.825 | 2.868 | 1,283 |
| *Participants who misreport* | | | |
| Boxes opened | 4.000 | 2.726 | 110 |
| Report minus first-box value | 4.455 | 2.384 | 110 |
| Maximum possible gain (9 - first box) | 6.591 | 1.988 | 110 |

*Notes:* Descriptive statistics. Columns report means, standard deviations, and sample sizes. The upper panel uses the full module sample for reported values and valid first-box observations for the remaining measures; the lower panel uses participants who report above the first-box value. The number of boxes opened counts distinct boxes.

Table A.3: Stopping after later openings

| | Dependent variable: *Stop* | | |
|---|---|---|---|
| | Unadjusted | Weighted | Fixed effects |
| | (1) | (2) | (3) |
| *New maximum* | 0.186*** | 0.199*** | 0.190*** |
| | (0.033) | (0.039) | (0.037) |
| N | 697 | 697 | 697 |
| Task-order control | No | No | Yes |
| Inverse-probability weighted | No | Yes | No |
| Risk-set fixed effects | No | No | Yes |
| Participants | 255 | 255 | 255 |

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

*Notes:* Opening-level OLS. Column (1) is unadjusted; Column (2) weights each opening by the inverse probability of its realized new-maximum outcome; Column (3) includes opening-position × prior-maximum fixed effects and controls for randomized task order. The sample includes reached third and later openings at which a new maximum is possible but not guaranteed. All columns include a constant. Standard errors clustered at the participant level in parentheses.

Table A.4: Final improvements and misreporting: full-sample specifications

| | Dependent variable: Misreport | |
|---|---|---|
| | (1) | (2) |
| *Final improvement* | 0.078*** | 0.045*** |
| | (0.010) | (0.013) |
| *Max gap* | | 0.016** |
| | | (0.008) |
| *Boxes opened* | | 0.005 |
| | | (0.008) |
| *Opened a second box* | | 0.067 |
| | | (0.043) |
| N | 1,283 | 1,283 |
| Task-order control | No | Yes |
| First-box fixed effects | No | Yes |
| $R^2$ | 0.164 | 0.229 |

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

*Notes:* Participant-level OLS. Column (1) includes only final improvement and a constant; Column (2) adds max gap, the number of distinct boxes opened, an indicator for opening a second box, first-box fixed effects, and a control for randomized task order. The outcome indicates a report above the truth. The sample includes all participants who opened at least one box. Final improvement is the increase in the running maximum at the last newly opened box and is zero after only one opening; max gap is the highest observed value minus the truth. Robust standard errors in parentheses.

# B Instructions

The text below is an English rendering of the German programmed questionnaire. It preserves the substantive participant-facing wording while regularizing minor translation artifacts. Interviewer instructions appear in *italics* and were not shown to participants. Square brackets describe text that the survey software inserted dynamically; internal programming variable names have been removed. The payoff tables reproduce the one-euro-per-reported-unit schedule. Task order was randomized; headings follow the module documentation.

### Intro

Now I would like to invite you to participate in a short and simple game. The task consists of two parts. What the two parts are about, you will learn immediately before. In both parts, you have a chance to win real money. But only one of the two parts will be selected for the final payout. You will flip a coin for it. If the coin shows "tails" after the toss, you will get the result for part 1. If the coin shows "heads", you will get the result for part 2. All information from the game, like all other information from this interview, will be evaluated anonymously only and will not be associated with your name. This is the end of the introduction.

*Interviewer: Please do not inform the target person in advance about the content of the two games! As the interviewer, make sure that the target person does not see the dice cup and dice before the dice game begins. Please do not give the dice cup and dice to the target person until the target person asks you for it!*

### Q490: Consent

Would you like to play the game? *Response options: Yes; No.*

*Interviewer: Before starting the module: As the interviewer, please try to make sure that no other interviewee is within earshot. As the interviewer, also try to leave the room or turn away so that the target person can answer the questions alone. It is important that the target person can complete the following tasks in complete privacy. If the target person agrees to play the game, please hand over the laptop to the target person for self-completion. After the experiment, the target person will be asked to hand over the laptop to you and continue the interview as regular.*

**Q491: Handover**

*Interviewer: Please hand over the laptop to the interviewee for self-completion.*

**Q493: Dice task**

The interviewer has just given you a 10-sided die and a dice cup. Now please roll the dice as many times as you like using the dice cup. Caution: Make sure that the interviewer does not see the result of your roll. Please remember the result of your first roll and enter it on the screen. How much you can earn in this part, if this part of the task is selected for payout, depends on the number you write down here. A higher number means a higher payout. You can see the exact payout for each reported number in the table below. The table will remain on the screen until you enter your number.

Note: Please have your interviewer hand you the dice cup and dice. Please make sure that no one can observe you playing the game.

| Reported number | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|
| Payoff (euros) | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |

**Q494: Box task**

In this part, you will see 10 black boxes on your screen. There are numbers behind the black boxes. The numbers behind the black boxes are 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9. The numbers are in random order, so the first box does not necessarily contain 1, the second does not necessarily contain 2, and so on. As soon as you click on a black box, the number hidden behind it is displayed. You can click on as many black boxes as you like. The laptop records how many and which boxes you click. Please remember the number behind the first black box you clicked and enter it on the screen. How much you earn in this part, if this part of the task is selected for payout, depends on the number you enter. A higher number means a higher payout. You can see the exact payout for each number reported in the table below. The table will remain on the screen until you enter your number.

| Reported number | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|
| Payoff (euros) | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |

**Q495: End of task**

You have reached the end of the task. Please hand the laptop back to your interviewer(s) and continue the survey as usual.

**Q496_0: Coin flip**

Now we will determine whether the first or second part of this task will be paid out. To do this, I ask you to flip a coin in a clearly visible position so that I can note down the result.

*Interviewer: Ask the target person to toss a coin in plain sight. Note whether "tails" or "heads" is on top.*

**Q496: Payment**

As you can see, [tails or heads] is on top. Therefore, you will now receive the payoff from [Part 1 or Part 2, as selected by the coin result]. You indicated that the first number in [the selected part] was [the participant's reported number]. Therefore, you will now receive [the reported number] euros from me.

*Interviewer: Please pay the target person the appropriate amount.*

## C A model of search and reporting

The model examines why participants might continue searching after they already know the truth and how the values they encounter can affect their final reports. We focus on search and reporting after the first opening, which establishes the truthful reporting benchmark. After the first opening, every report remains available, and further search changes neither the truth nor the payment schedule. We first establish a benchmark in which additional exposure leaves reporting utility unchanged, so further opening only adds costs. We then allow encountering a number above the truth to increase the utility of reporting that number, capturing the possibility that personal exposure makes false reporting easier to justify. This exposure benefit can make further search worthwhile. We study the resulting opening, stopping, and reporting decisions and derive the conditions supporting the three predictions in Section 3.

**Setup.** Ten boxes contain each value in $X = \{0, \ldots, 9\}$ exactly once, assigned uniformly at random across boxes. Participants know this and are instructed to report the value in the first box they open. Every report $r \in X$ remains available, including before any box has been opened. The $k$th distinct opening has a known cost $c_k > 0$ in utility units.[9] For a given participant type, costs may vary with opening position but do not depend on the values in the boxes. Reporting without opening any box entails an additional psychological cost. This cost applies only when no box has been opened and may be smaller or larger than the cost of the first opening. The initial decision depends on these costs and on reporting utility both with and without personal observation. Opening the first box makes truthful reporting possible, although false reports may still carry psychological costs.

We study subsequent search and reporting conditional on the first opening. Its value $x_1$ is therefore known and provides the truthful reporting benchmark. Let $E \subseteq X$, with $x_1 \in E$, denote the set of personally revealed values. Each additional opening reveals a value drawn uniformly from $X \setminus E$. Participants know their preferences, correctly

[9] Opening costs may reflect the time, attention, or effort required to reveal and process a value, even though opening carries no monetary charge. They create a tradeoff between the potential reporting benefit of further exposure and the cost of continuing search. When exposure provides no reporting benefit, positive costs make further opening strictly unattractive.

remember the truth and revealed values, and gain nothing from further opening beyond its effect on reporting utility, $U(r; x_1, E)$. They maximize expected reporting utility net of additional opening costs and stop when indifferent between stopping and continuing.

**Proposition 1** (No further search)**.** *Conditional on having opened the first box, if $U(r; x_1, E)$ does not depend on $E$, the participant opens no additional boxes.*

**Proof.** Every subsequent search ends with the same best reporting utility,

$$\max_{r \in X} U(r; x_1, \{x_1\}),$$

which stopping immediately attains. Any strategy that opens additional boxes with positive probability adds positive expected costs. The first-opening cost is already incurred. □

**Personal exposure.** We allow personally encountering a number above the truth to make reporting that number more attractive. Let $u(r; x_1)$ denote baseline reporting utility, incorporating the payment and any psychological cost of incorrect reporting. We measure utility relative to truthful reporting, so $u(x_1; x_1) = 0$. Personally revealing a value $r > x_1$ adds a benefit $\delta(r; x_1) \geq 0$, capturing the possibility that reporting a number one has seen is easier to justify:

$$U(r; x_1, E) = u(r; x_1) + \delta(r; x_1)\mathbf{1}\{r \in E\}, \qquad \delta(r; x_1) \geq 0, \quad \delta(r; x_1) = 0 \text{ for } r \leq x_1. \tag{1}$$

The indicator $\mathbf{1}\{r \in E\}$ equals one when the reported value has been personally revealed and zero otherwise. Revealing a value therefore changes only the utility of reporting that value. The exposure benefit applies only to misreports. Knowing that a number remains in the boxes, or even inferring its location, does not generate the benefit; the participant must personally reveal it.

If the participant stops with revealed set $E$, they choose the report that gives the highest reporting utility. The value of stopping is therefore

$$W(E; x_1) = \max_{r \in X} U(r; x_1, E).$$

Alternatively, the participant can pay $c_{|E|+1}$ to open another box. Each remaining value is equally likely to be revealed, and afterward the participant can again choose whether to stop or continue.

Let $V(E; x_1)$ denote the best expected utility attainable from the current state, net of future opening costs. It satisfies the following Bellman equation, which compares stopping now with opening another box:

$$V(E; x_1) = \max\Big\{W(E; x_1),\ -c_{|E|+1} + \frac{1}{|X \setminus E|} \sum_{v \in X \setminus E} V(E \cup \{v\}; x_1)\Big\}, \tag{2}$$

for $E \neq X$. The first term is the value of stopping now; the second is the expected value of another opening, including the option to search further afterward. Once all values have been revealed, no further opening is possible, so $V(X; x_1) = W(X; x_1)$.

**The first value.** Assume that truthful reporting is weakly preferred to every other report before additional exposure, so $u(r; x_1) \leq 0$ for every $r$. For an observed value above the truth with reporting gap $d = r - x_1 > 0$, define

$$g(d; x_1) = u(x_1 + d; x_1) + \delta(x_1 + d; x_1).$$

This is the utility of reporting that value relative to truthful reporting. Assume that $g(d; x_1)$ is weakly decreasing in $x_1$ for each feasible fixed gap $d$. Thus, an observed lie of the same size is no more attractive relative to truth telling when the first value is higher.

Stopping after the first opening yields $W(\{x_1\}; x_1) = 0$. The net value of opening a second box and making subsequent decisions optimally is therefore

$$C(x_1) = -c_2 + \frac{1}{9} \sum_{v \neq x_1} V(\{x_1, v\}; x_1).$$

The participant opens a second box exactly when $C(x_1) > 0$.

**Proposition 2** (A higher first value discourages opening a second box)**.** *Under the exposure specification in Equation (1), suppose that $u(r; x_1) \leq 0$ for every report $r$ and first value $x_1$, and that $g(d; x_1)$ is weakly decreasing in $x_1$ for each fixed positive gap $d$ over the first values for which that gap is feasible. Then $C(x_1)$ is weakly decreasing in $x_1$.*

**Proof.** Let $h(d; x_1) = \max\{0, g(d; x_1)\}$. Since $u \leq 0$, stopping utility equals the largest revealed gain, or zero if none is positive. At first value $x_1$, the nine remaining boxes offer one gain $h(d; x_1)$ for each positive gap $d = 1, \ldots, 9 - x_1$, together with $x_1$ zero gains. When $x_1 = 9$, all nine gains are zero.

For $x_1 \leq 8$, compare this problem with first value $x_1 + 1$. Pair gains corresponding to the same positive gap $d \leq 8 - x_1$, and pair the $x_1$ zero gains. Pair the remaining

gain $h(9 - x_1; x_1)$ with the additional zero at the higher first value. By the assumed monotonicity of $g$ and the nonnegativity of $h$, every gain at the lower first value weakly exceeds its partner.

This pairing induces equally likely matched opening orders. At every matched state, stopping utility is weakly higher at the lower first value, while opening positions and costs coincide. Backward induction on Equation (2) therefore gives weakly higher optimal value at the lower first value. Since the second-opening cost is also identical, averaging over matched second draws yields $C(x_1) \geq C(x_1 + 1)$. □

Because the first value is randomly assigned, the individual prediction also implies that the average probability of opening a second box is weakly decreasing in the first value, provided the assumptions hold for all participant types.

**New maxima and stopping.** Suppose that larger observed values above the truth offer weakly greater reporting gains, so $[g(d; x_1)]_+ = \max\{0, g(d; x_1)\}$ is weakly increasing in $d$. Under $u \leq 0$, this condition is equivalent to the best available reporting utility depending only on the highest value seen and being weakly increasing in that value. We can therefore write $W(E; x_1) = w(\max E)$, with $w$ weakly increasing. A draw below the current maximum leaves this utility unchanged, while a new maximum may increase it.

Consider the same participant at the same point in search, with the same values already revealed. The following proposition compares their stopping decision if the next box reveals a lower number with their decision if it reveals a higher one.

**Proposition 3** (Higher draws weakly favor stopping)**.** *Fix $x_1$ and suppose $W(E; x_1) = w(\max E)$ for every $E \ni x_1$, with $w$ weakly increasing. From the same revealed set $E$, consider two possible next draws, $a < b$, both currently unopened. If the participant would stop after revealing $a$, they would also stop after revealing $b$. In particular, if they would stop after a draw below the current maximum, they would also stop after any draw that sets a new maximum.*

**Proof.** For a state $F$ with unopened boxes, let $D(F)$ be the largest expected gain relative to stopping now, net of future opening costs, among strategies that open at least one more box. Stopping is optimal exactly when $D(F) \leq 0$.

Write $A = \max\{\max E, a\}$ and $B = \max\{\max E, b\}$, so $A \leq B$. Take any strategy that opens at least one more box from $E \cup \{b\}$. Starting instead from $E \cup \{a\}$, the participant

can follow the same search and stopping rule by interpreting a subsequent draw of $b$ as a draw of $a$ and leaving every other draw unchanged. This pairs the uniformly random opening orders, with the same costs at the same opening positions.

Let $R$ be the highest value in $E$ or among the subsequent draws on the higher-draw path, excluding $a$. The higher-draw path ends with maximum $\max\{B, R\}$, while the paired lower-draw path ends with a maximum of at least $\max\{A, R\}$. The reporting gains satisfy

$$w(\max\{A, R\}) - w(A) \;\geq\; w(\max\{B, R\}) - w(B).$$

To see why, the gain from reaching $R$ from a starting maximum $t$ is $\max\{0, w(R) - w(t)\}$, which is weakly decreasing in $t$.

The paired strategy therefore yields at least as much additional reporting utility at the same cost. Maximizing over strategies gives

$$D(E \cup \{a\}) \geq D(E \cup \{b\}).$$

If $D(E \cup \{a\}) \leq 0$, then $D(E \cup \{b\}) \leq 0$ as well. Thus, stopping after $a$ implies stopping after $b$. □

Among participants who reach an opening, the probability of a new maximum depends only on the opening number and previous maximum. If the stopping proposition's assumptions hold for every participant type, it therefore predicts a weakly higher average probability of stopping after a new maximum than after a lower draw within cells defined by these two variables.

**Reports that match observed values.** We now ask which misreports can be optimal. Suppose that, without an exposure benefit, every misreport is strictly less attractive than truthful reporting. Personal exposure can offset this disadvantage for values the participant has actually revealed.

**Proposition 4** (Misreports match observed values)**.** *If $u(r; x_1) < 0$ for every $r > x_1$, any optimal misreport matches a value in $E$.*

**Proof.** For $r > x_1$ with $r \notin E$, Equation (1) gives $U(r; x_1, E) = u(r; x_1) < 0 = U(x_1; x_1, E)$, so truth telling is strictly better. □

Under this strict restriction, weakly increasing $g(d; x_1)$ makes the observed maximum optimal whenever a misreport is optimal. A flat $g$ allows smaller observed values above the truth to tie; if $g$ declines at larger gaps, the maximum need not be optimal.

# D Additional analyses

## D.1 Dice-task comparison

This appendix examines whether reports in the private dice task are related to how participants search and report in the box task. Because we do not observe the actual first roll, a high report could reflect either a lucky roll or misreporting. We define *Dice excess* as the reported die roll minus 4.5, the expected value of a fair ten-sided die. This measure cannot tell us whether an individual participant lied, but it allows us to examine whether participants who report higher rolls also search and report differently in the box task.

Table D.1 reports linear probability models in Columns 1–5 and an OLS regression for the number of boxes opened in Column 6. In Columns 1–2, the dependent variable equals one if the participant misreports, defined as reporting above the truth, and zero otherwise. Columns 3–4 use one observation per opening of a new box after the first; the dependent variable equals one if the participant opens no further new box, and zero otherwise. In Column 5, the dependent variable indicates whether the participant opens a second distinct box. In Column 6, it is the total number of distinct boxes opened, including the first. Columns 1, 2, 5, and 6 use one observation per participant.

Column 1 includes only dice excess and a constant. Columns 2–6 include first-box fixed effects and control for randomized task order. Column 2 also includes *Final improvement* (the increase in the maximum at the last newly opened box), *Max gap* (the highest observed value minus the truth), the number of distinct boxes opened, and an indicator for opening a second box. Columns 3–4 include *Current improvement* and additional fixed effects for opening position, the current number, and the previous maximum. Column 4 adds the interaction between *Current improvement* and dice excess.

The results show that higher dice reports are associated with a greater probability of misreporting in the box task (Columns 1–2). Adding dice excess leaves the association between *Final improvement* and misreporting similar to that reported in Column 2 of Table A.4 (Column 2 of Table D.1). Column 3 shows that larger *Current improvements* are associated with a greater probability of stopping after controlling for dice reports, consistent with the stopping response to new maxima in Panel B of Table 2. We do not find any clear association between higher reported dice rolls and stopping, opening a second box, or the total number of distinct boxes opened (Columns 3, 5, and 6).

Column 4 examines whether the stopping response to improvements varies with dice reports by including an interaction between *Current improvement* and *Dice excess*. The positive interaction suggests that larger improvements in the highest number seen are associated with a sharper increase in stopping probability among participants who report higher numbers in the dice task. This pattern is consistent with the idea that the search path helps participants act on a willingness to misreport by providing a favorable number they can more easily justify reporting. Because some participants completed the dice task after the box task, this pooled comparison cannot establish whether that willingness preceded their box-task behavior.

Table D.1: Dice-task reports and box-task behavior

| | (1) Misreport | (2) Misreport | (3) Stops after opening | (4) Stops after opening | (5) Opens second box | (6) Boxes opened |
|---|---|---|---|---|---|---|
| *Dice excess* | 0.007** | 0.005** | 0.001 | -0.004 | 0.004 | 0.000 |
| | (0.003) | (0.002) | (0.004) | (0.004) | (0.005) | (0.023) |
| *Final improvement* | | 0.044*** | | | | |
| | | (0.013) | | | | |
| *Current improvement* | | | 0.067*** | 0.061*** | | |
| | | | (0.012) | (0.012) | | |
| *Current improvement* × *Dice excess* | | | | 0.005** | | |
| | | | | (0.002) | | |
| N | 1283 | 1283 | 1489 | 1489 | 1283 | 1283 |
| $R^2$ | 0.004 | 0.231 | 0.176 | 0.179 | 0.049 | 0.025 |
| Fixed effects | None | First-box | Full | Full | First-box | First-box |
| Task-order control | No | Yes | Yes | Yes | Yes | Yes |
| Outcome mean | 0.086 | 0.086 | 0.263 | 0.263 | 0.306 | 2.161 |

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

*Notes:* OLS regressions. Columns (1)–(2) use misreporting as the outcome; Columns (3)–(4) use stopping after a post-first opening; Column (5) uses opening a second box; Column (6) uses the number of distinct boxes opened. Columns (1), (2), (5), and (6) have one observation per participant; Columns (3)–(4) have one per post-first opening. Column (1) includes only dice excess and a constant. Columns (2)–(6) include first-box fixed effects and control for randomized task order. Column (2) also controls for max gap, the number of distinct boxes opened, and opening a second box. Columns (3)–(4) additionally include opening-position, current-value, and prior-maximum fixed effects; Column (4) adds the displayed interaction. All columns include a constant. Dice excess is the reported die value minus 4.5; max gap is the highest observed value minus the truth. Standard errors are robust in Columns (1), (2), (5), and (6) and clustered at the participant level in Columns (3)–(4).

## D.2 Memory and reporting benchmarks

### Recall conditional on an incorrect report

We examine whether simple recall rules can explain which numbers participants report when they give an incorrect answer. Table D.2 focuses on the 115 participants who opened additional boxes and reported a number different from the truth. For each participant, we hold the actual opening sequence fixed and consider reports drawn from the numbers in the additional boxes. We exclude the truthful first number because this exercise conditions on an incorrect report.

We consider three recall rules. Uniform recall gives each of these numbers an equal chance of being reported. Linear recency makes numbers encountered later more likely to be recalled, assigning weights of 1 to the first additional opening, 2 to the second, and so on. Payoff-increasing recall makes higher numbers more likely to be recalled, assigning weight $v+1$ to a number $v$. For each rule, we calculate the expected number of misreports and reports matching the highest number seen. We then use 50,000 simulations to assess each count separately against the 96 misreports and 68 maximum matches observed in the data.

Giving each number an equal chance, or favoring more recent numbers, predicts significantly fewer misreports and maximum matches than observed. Favoring higher numbers brings both predictions closer to the data, and neither comparison rejects this rule. The exercise therefore leaves open the possibility that participants remember high numbers more readily.

Table D.2: Recall benchmarks conditional on an incorrect report

| Recall rule | Misreport | $p$ | Maximum match | $p$ |
|---|---|---|---|---|
| Observed | 96 | | 68 | |
| Uniform | 82.6 | $< 0.001$ | 53.9 | $< 0.001$ |
| Payoff-increasing | 93.7 | 0.541 | 65.8 | 0.659 |
| Linear recency | 86.0 | 0.003 | 58.9 | 0.022 |

*Notes:* Conditional recall benchmarks. Columns report counts of misreports and maximum matches, each with a separate $p$-value. The sample comprises participants who opened additional boxes and reported incorrectly. Each benchmark holds actual search paths fixed and draws from distinct post-first values, excluding the truth. Uniform recall assigns equal weights, payoff-increasing recall assigns $v + 1$, and linear recency assigns post-first opening ranks. Expected counts sum participant-specific probabilities. Two-sided $p$-values use 50,000 simulation draws.

### Intended truth telling, recall, and reporting slips

We next examine whether a combination of truthful reporting, recall, and reporting mistakes can explain the broader reporting pattern. Table D.3 includes all 392 participants who opened additional boxes. In the model, a participant may report the truthful first number, recall a number from any of the distinct boxes opened, or make a reporting slip that selects each number from 0 to 9 with equal probability. Recall now includes the first-box number, so recalling an opened number can also produce a truthful report.

For each recall rule, we set how often the model uses truthful reporting, recall, or a reporting slip so that it predicts the observed totals of 115 incorrect reports and 13 reports of unseen numbers. We use the same probabilities for everyone. Since the model is designed to match these totals, we assess whether it also reproduces other reporting patterns. When recall favors higher numbers, for example, the model predicts 43.4 reports matching the last newly opened value, compared with 74 observed. It can therefore match how often participants report incorrectly while still underestimating how often they report the number in the last newly opened box.

We then test whether each model can reproduce three counts together: misreports, reports matching the highest number seen, and reports matching the last newly opened value. For each model, we generate 10,000 simulated datasets, keeping participants' actual opening paths fixed and re-estimating the probabilities of the three processes in every simulation. All three models are rejected at $p < 0.001$. These particular combinations

of truthful reporting, recall, and slips therefore fail to reproduce the observed reporting patterns. However, we assume the same mix of truthful reporting, recall, and slips for everyone and keep each participant's actual search path fixed. The tests therefore leave open explanations in which some people are more prone to errors than others, certain numbers are easier to remember than these rules allow, or what people remember affects how they search and when they stop.

Table D.3: Intended truth telling, recall, and reporting slips

| Recall rule | Misreport | Downward | Maximum match | Last-new match | Joint $p$ |
|---|---|---|---|---|---|
| Observed | 96 | 19 | 129 | 74 | |
| Uniform | 66.5 | 48.5 | 79.9 | 37.9 | $< 0.001$ |
| Payoff-increasing | 81.9 | 33.1 | 98.7 | 43.4 | $< 0.001$ |
| Linear recency | 68.3 | 46.7 | 82.9 | 47.0 | $< 0.001$ |

*Notes:* Fitted reporting mixtures. Columns report observed and expected reporting counts and joint $p$-values. The sample comprises all participants who opened additional boxes. Each model combines intended truth telling, recall from distinct opened values including the truth, and reporting slips uniform over 0–9. Recall weights are equal, $v + 1$, or distinct-opening ranks, as labelled. Common nonnegative component probabilities sum to one and are fitted to the incorrect- and unseen-report counts. Maximum and last-new matches include all reports, regardless of direction. The joint statistic is the largest absolute standardized residual across misreport, maximum-match, and last-new-match counts. Its $p$-value uses 10,000 parametric bootstrap draws with refitting and a plus-one correction. Actual paths are fixed; reporting draws are independent across participants.